

CALCULUS WITH

Asante

TOPIC ONE | TOPIC SERIES

Series, Limits of a
Series & p-Series

Compiled and taught by Asante

MATH 152 | Calculus & Analysis | KNUST BMEI

Why This Booklet Exists

This booklet is here to supplement your own learning — not replace it. Lectures, your notes, your textbook, and the time you spend wrestling with a problem on your own all still matter. Nothing here is a shortcut around doing the work; it's a second angle on the same material, laid out the way I wish someone had laid it out for me.

Learning is not a straight line. You will understand the p-series test on page 9, forget it exists two weeks later, and only really get it when it shows up again in a totally different problem. That's normal. This booklet is built to be revisited, not read once and shelved.

If you get stuck, that's the point where the real learning starts, *chale* — not a sign that something is wrong with you. Sit with the confusion for a bit before you flip to the answer.

Every booklet in this series also closes with a Cheat Sheet — a fast-reference summary of the whole topic. It's there for review the night before a test, not for learning something for the first time.

A Note on How to Use This

Don't try to read this cover to cover in one sitting the night before a test. Work through one section, attempt the boxed questions honestly, and come back to whatever didn't stick.

How This Booklet Works

Every topic in this series is laid out the same way, so once you learn the pattern once, you can move faster through every booklet after this one. Here's what each marker means.

TRY THIS FIRST

A few questions up front, before any explanation. Attempt them cold — they tell you exactly which part of this page you actually need.

DEFINITION

The formal statement — rule, theorem, or term — stated precisely. Worth memorising word for word.

EXAMPLE

A worked problem, solved in full, showing the reasoning step by step, not just the answer.

TRY IT

Your turn. Attempt it before checking the answer given underneath.

RECALL / STRATEGY

A shortcut, common trap, or the exact test to reach for — the thing you should be thinking right before you attempt a problem.

WATCH & LEARN

Hyperlinked videos, dropped in right next to the concept they explain — click (▶) whenever reading alone isn't landing.

Each booklet ends with a Problem Set and a Cheat Sheet summary. Questions

marked **T** are technology-appropriate — a calculator or CAS is fine for checking arithmetic, not for skipping the reasoning. Full worked solutions live in the separate Answer Key that accompanies this booklet.

Asante Inc. understands that learning is not a linear process and can be stressful — so this booklet has hyperlinked videos built in throughout, plus a short list of videos worth watching before you even start reading. Look for ► next to any concept that isn't landing.

Key Vocabulary

A quick glossary before you dive in. Come back here any time a term stops making sense mid-page.

Sequence

An ordered list of numbers, $a_1, a_2, a_3, \dots$, one after another. Think of it like a savings account balance checked at the end of every month — one number per month, in order.

Series

What you get when you add up the terms of a sequence. If the sequence is monthly deposits, the series is your running total balance over time.

Partial Sum (S_n)

The running total after adding the first n terms of a series. It's the balance snapshot at one particular point — not the final answer, just where things stand after n steps.

Convergence

A series converges when its partial sums settle down and approach one fixed number as you add more and more terms — like a phone battery percentage easing toward 100% and staying there, rather than climbing forever.

Key Vocabulary (continued)

Divergence

A series diverges when its partial sums do not settle to a single number — they grow without bound, oscillate, or otherwise refuse to land anywhere.

n -th Term

The general formula for the term in position n of a sequence, written a_n . It's the “recipe” that tells you what the 5th, 50th, or 500th term is without listing every term before it.

Recommended Videos --- Watch Before Reading

Just these five before you start — they carry the backbone ideas the rest of the booklet builds on.

- ▶ [Intro to Sequences](#)
- ▶ [Definition of a Limit](#)
- ▶ [Converging and Diverging Sequences Using Limits](#)
- ▶ [Monotonic Sequences and Boundedness](#)
- ▶ [How to Tell if Any Sequence Is Convergent or Divergent](#)

Every video used across this booklet — and every video still to come for future topics — lives in one running playlist:

- ▶ [Full Playlist — Calc with Analysis](#)

Try This First --- 5 Questions, No Notes

1. Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge or diverge?
 - (A) Converges to 1
 - (B) Diverges
 - (C) Converges to 0
 - (D) Cannot be determined

2. Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge?
 - (A) Diverges
 - (B) Converges to 0
 - (C) Cannot be determined
 - (D) Converges

3. Find $\lim_{n \rightarrow \infty} \frac{2n+1}{3n-5}$.
 - (A) $\frac{2}{3}$
 - (B) 0
 - (C) 1
 - (D) Does not exist

4. For which values of p does $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?
 - (A) $p < 1$
 - (B) $p \leq 1$
 - (C) $p > 1$
 - (D) All p

5. Does $\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge or diverge?
 - (A) Converges to 1

- (B) Converges to 0
- (C) Cannot be determined by the n -th term test
- (D) Diverges

HOW DID THAT FEEL?

If all five came easily, skim the boxed rules below and jump to the Problem Set. If any of them made you pause — especially #1 vs. #2, which look almost identical — stick around. That's exactly the trap this topic is built around, and by the end of these few pages it won't catch you again.

1. What a Series Actually Is

DEFINITION --- SEQUENCE VS. SERIES

A sequence (a_n) is just a list: $a_1, a_2, a_3, \dots$

A series is what you get when you add the terms of a sequence:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots$$

The n -th partial sum is $S_n = a_1 + a_2 + \dots + a_n$ — the running total after n terms.

DEFINITION --- CONVERGENCE OF A SERIES

The series $\sum a_n$ converges to L if the sequence of partial sums has a finite limit:

$$\lim_{n \rightarrow \infty} S_n = L$$

If this limit doesn't exist (or is $\pm\infty$), the series diverges.

This is the one idea the rest of this topic hangs on: a series converges when its running total settles down to one fixed number, no matter how far out you go.

► [Watch: Intro to Sequences](#)

2. The Trap: “The Terms Shrink” Is Not Enough

RECALL / STRATEGY --- THE n -TH TERM TEST

If $\lim_{n \rightarrow \infty} a_n \neq 0$, the series must diverge. This is a quick first check.

But the converse is false: $a_n \rightarrow 0$ does *not* guarantee convergence. The harmonic series below is the classic counterexample — keep it in your back pocket.

Example 1 --- The Harmonic Series

Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge?

Solution. The terms $\frac{1}{n} \rightarrow 0$, so the n -th term test doesn't rule it out. But grouping terms shows the partial sums grow without bound:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{>1/2} + \underbrace{\left(\frac{1}{5} + \cdots + \frac{1}{8}\right)}_{>1/2} + \cdots$$

Each bracketed group exceeds $\frac{1}{2}$, so $S_n \rightarrow \infty$. The harmonic series diverges, even though its terms shrink to 0.

► Watch: Definition of a Limit

3. p-Series: One Test, One Number to Check

DEFINITION --- p-SERIES

A series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is called a p-series. Its behaviour depends entirely on the exponent p .

RECALL / STRATEGY --- THE p-SERIES TEST

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$.

The harmonic series is the boundary case $p = 1$ — which is exactly why it diverges despite its terms going to 0.

Example 2 --- Applying the p-Series Test

Determine convergence: (a) $\sum \frac{1}{n^3}$ (b) $\sum \frac{1}{\sqrt{n}}$

Solution. (a) Here $p = 3 > 1$, so the series converges.

(b) Rewrite $\frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$, so $p = \frac{1}{2} \leq 1$. The series diverges.

TRY IT --- YOUR TURN

Classify each as converging or diverging.

(a) $\sum \frac{1}{n^4}$ (b) $\sum \frac{1}{n^{0.9}}$ (c) $\sum \frac{1}{n^{1.5}}$

Answers: (a) converges, $p = 4 > 1$ (b) diverges, $p = 0.9 \leq 1$ (c) converges, $p = 1.5 > 1$

► Watch: Converging and Diverging Sequences Using Limits

4. Limits of a Sequence --- The Building Block

Before a series can converge, we often need the limit of the sequence itself. This is ordinary limit-at-infinity work: compare growth rates of numerator and denominator.

Example 3 --- Limit of a Rational Sequence

Find $\lim_{n \rightarrow \infty} \frac{5n^2 + 3}{2n^2 - 1}$.

Solution. Divide top and bottom by the highest power, n^2 :

$$\lim_{n \rightarrow \infty} \frac{5 + \frac{3}{n^2}}{2 - \frac{1}{n^2}} = \frac{5 + 0}{2 - 0} = \frac{5}{2}$$

TRY IT --- YOUR TURN

Find $\lim_{n \rightarrow \infty} \frac{n + 1}{n^2 + 1}$.

Answer: 0 — the denominator grows faster, so the fraction is squeezed to 0.

► Watch: Monotonic Sequences and Boundedness

► Watch: Bounded, Monotone Sequences & Limits

5. Putting the n -th Term Test to Work

Example 4 --- Diverging by the n -th Term Test

Does $\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge?

Solution. Check the limit of the terms first, before reaching for any heavier machinery:

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1 \neq 0$$

Since the terms don't shrink to 0, the series diverges — no partial-sum analysis needed. This is the fastest tool in the kit; always reach for it first.

WORTH TRYING ON YOUR OWN

Plot the partial sums S_n for the harmonic series and for $\sum 1/n^2$ side by side (a spreadsheet or calculator works fine, n up to 500). One climbs forever, however slowly; the other visibly flattens toward a ceiling — the fastest way to make “converges vs. diverges” stop feeling abstract.

► Watch: How to Tell if Any Sequence Is Convergent or Divergent

Putting It Together

- A series converges iff its partial sums S_n approach a single finite limit.
- $a_n \rightarrow 0$ is necessary but never sufficient for convergence — the harmonic series is the proof.
- p-series $\sum 1/n^p$: converges for $p > 1$, diverges for $p \leq 1$. One number, one decision.
- Sequence limits at infinity: divide through by the highest power before taking the limit.

Problem Set Questions marked **T** are technology-appropriate — a calculator or CAS is fine for checking your arithmetic, not for skipping the reasoning.

1. Does $\sum \frac{1}{n^5}$ converge or diverge?
 - (A) Diverges
 - (B) Cannot be determined
 - (C) Converges
 - (D) Converges to 0
2. Does $\sum \frac{1}{n^{0.5}}$ converge or diverge?
 - (A) Diverges
 - (B) Converges
 - (C) Converges to 0
 - (D) Cannot be determined
3. Does $\sum \frac{1}{n}$ converge or diverge, and why does the n -th term test fail to catch this?
 - (A) Converges; the n -th term test confirms it since $a_n \rightarrow 0$
 - (B) Diverges; the n -th term test is inconclusive here since $a_n \rightarrow 0$, which doesn't guarantee convergence
 - (C) Diverges; the n -th term test proves divergence directly
 - (D) Cannot be determined without more advanced tests
4. For what values of p does $\sum \frac{1}{n^p}$ diverge?
 - (A) $p \leq 1$
 - (B) $p < 1$
 - (C) $p > 1$
 - (D) All real p
5. Find $\lim_{n \rightarrow \infty} \frac{7n - 2}{4n + 9}$.
 - (A) 0
 - (B) $\frac{7}{4}$

- (C) 1
(D) Does not exist
6. Find $\lim_{n \rightarrow \infty} \frac{3n^2 + n}{n^3 + 1}$.
- (A) 3
(B) 1
(C) Does not exist
(D) 0
7. **T** Use the n -th term test to show $\sum \frac{n}{n+1}$ diverges.
- (A) $\lim a_n = 0$, so the series converges
(B) $\lim a_n$ does not exist, so divergence is inconclusive
(C) $\lim a_n = 1 \neq 0$, so the series diverges
(D) $\lim a_n = 1$, so the series converges
8. Classify $\sum \frac{1}{n^{1.01}}$ and explain, in one sentence, why it sits so close to the boundary case.
- (A) Diverges; $p = 1.01$ is basically the harmonic series
(B) Converges; $p = 1.01 > 1$, just barely past the boundary $p = 1$
(C) Cannot be determined without numeric approximation
(D) Converges; because 1.01 rounds to 1